



Vantage® Research Ultrasound System

Vantage Tool Summary

D00035 Rev. A

Summary

This document provides instructions on how to utilize the Vantage System tools and utilities.

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1. Introduction

A number of useful tools and utilities can be found in the *Tools* subfolder within the Vantage project folder. An overview of each of these tools is listed here. In addition, demonstration videos are available on the Verasonics website: <http://verasonics.com/tools-demonstration/>.

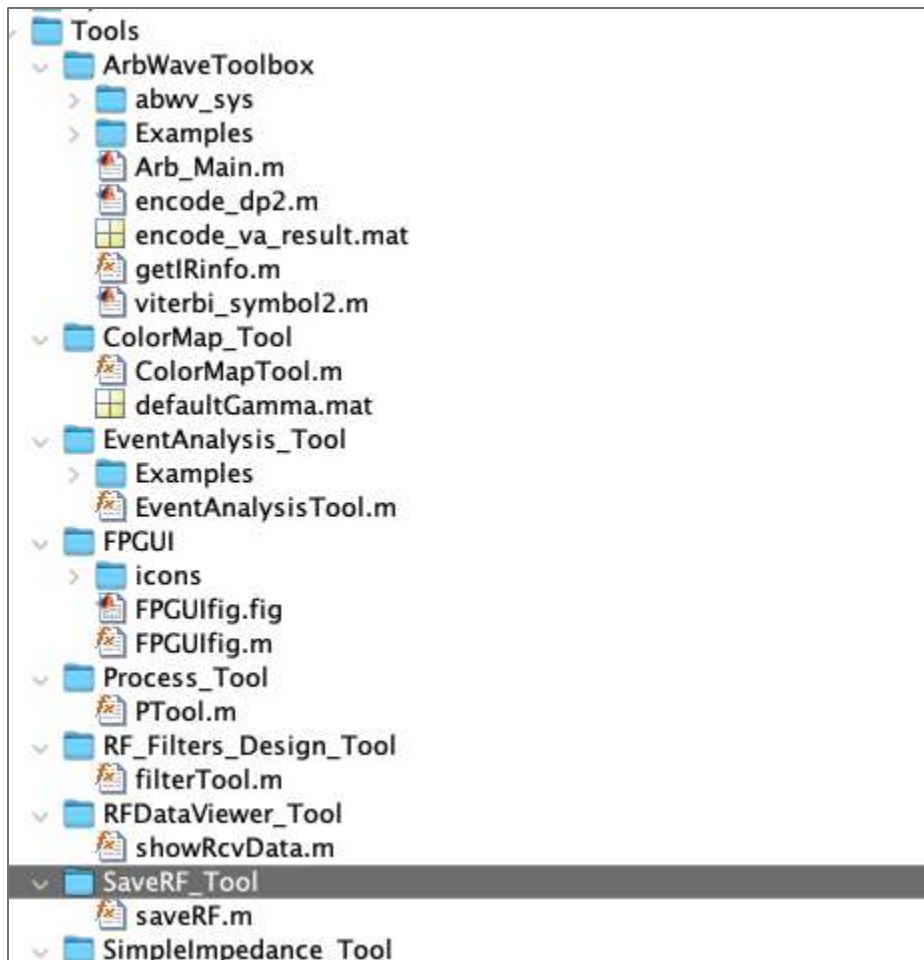


Figure 1: Tools and Utilities in the Tools Subfolder

Refer to the [Vantage Release Notes](#) for any updates or known issues associated with the tools described in this document.

2. Arbitrary Waveform Tool

ArbWaveToolbox is a collection of utilities and examples to help define and analyze waveforms using the Vantage Research Ultrasound System's arbitrary waveform transmit features. Documentation on how to use these utilities is included within the *ArbWaveToolbox* subfolder of the *Tools* directory. A video is provided on the Verasonics website.

3. Color Map Design Tool

From a clinical standpoint, a color map enables the eyes to perceive the subtle variations in tissue texture. The Color Map Tool can be used to adjust the gamma curve of the display window and to visualize the effect on the image in real-time. The users have the option to save up to five curve plot settings for viewing different tissues.

After running a script in simulation mode, the user has the option to change the contrast of the display window using the Color Map tool. The user can utilize the Color Map tool in the following ways:

- Select one of the default Gamma curves for setting the contrast of the display window.
- Customize a Gamma curve by tweaking one of the default Gamma curve plots.

Use the following procedure for utilizing the Color Map tool:

1. Execute a script in simulation mode before using the Color Map tool—the VSX Control GUI appears.
2. In the VSX Control GUI, choose the ColorMapTool option from the Tools drop-down menu.

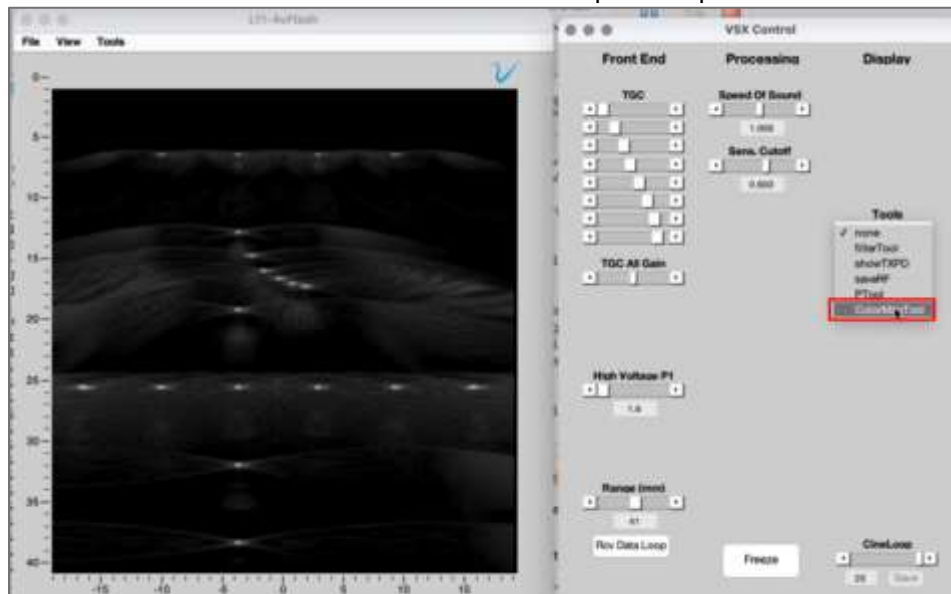


Figure 2: Color Map tool selection using the VSX Control GUI

3. Observe that the Color Map tool appears as a pop-up window.

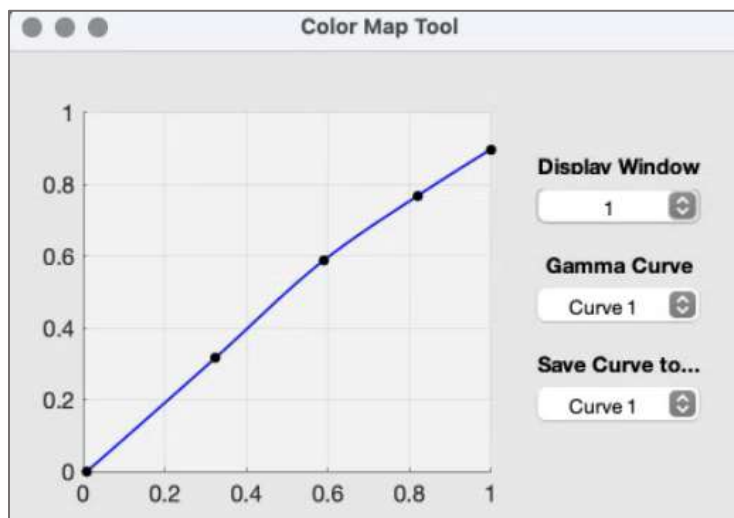


Figure 3: Color Map tool GUI displaying a gamma curve

4. From the Display Window drop-down menu, select the number of display windows that need to be adjusted. For example, in Figure 4, the number of display windows is 1.
5. From the Gamma Curve drop-down menu, select a gamma curve option. The Gamma Curve drop-down menu has five default gamma curve options available. The selected Gamma curve option is applied immediately in real-time to the display window.
6. After selecting a Gamma curve, tweak the selected Gamma curve to optimize for the contrast of the display window. To do this, click and hold the left mouse button on the Gamma curve plot and drag the mouse to adjust as appropriate.

- After customizing the Gamma curve, use the 'Save Curve to...' drop-down menu to assign a name for the saved option. As an example, the user can select Gamma Curve 4 from the Gamma Curve drop-down menu, tweak the Gamma plot and save it as Gamma Curve 1 using the 'Save Curve to...' drop-down menu. This Gamma curve will now be saved as Gamma Curve 1 in the Gamma Curve dropdown menu.

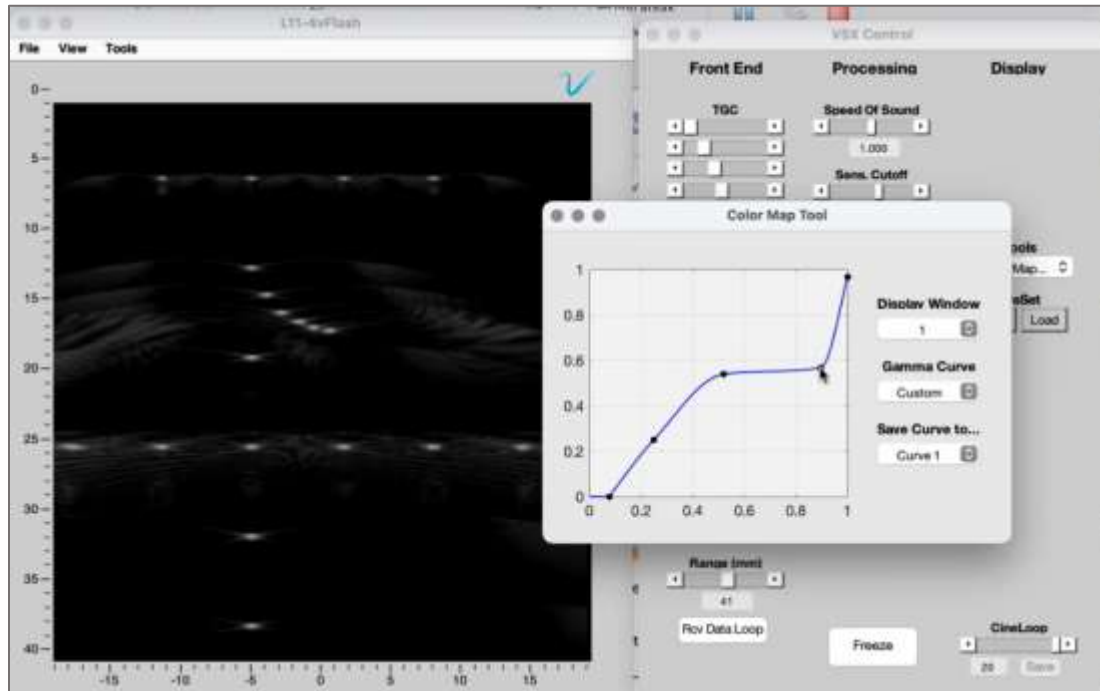


Figure 4: Customize a Gamma curve

4. Event Analysis Tool

The Event Analysis Tool assists with developing and debugging a setup script.

After running a SetUp script, the user can view the output in the MATLAB® workspace by entering the `EventAnalysisTool` at the MATLAB command prompt. Here, the user sees a visual display of all events in the event sequence.

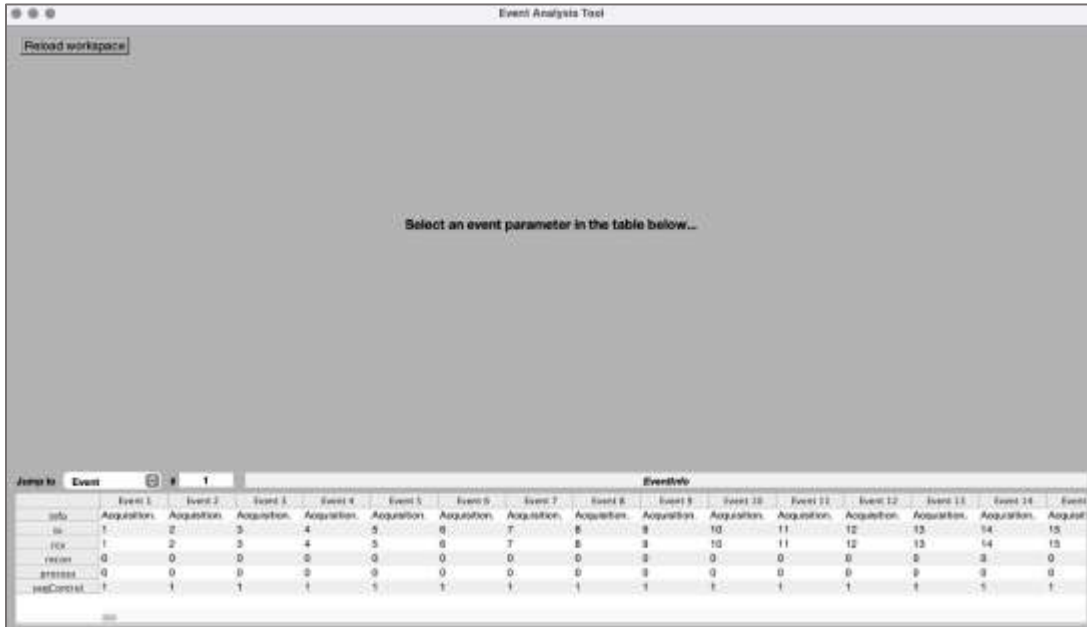


Figure 5: Event Analysis Tool

By clicking on any Event, the user can see a display of the content of structures indexed by that Event. This makes it easy to find and fix errors in indexing logic and helps the user visualize system behavior as they step through each event in the event sequence.

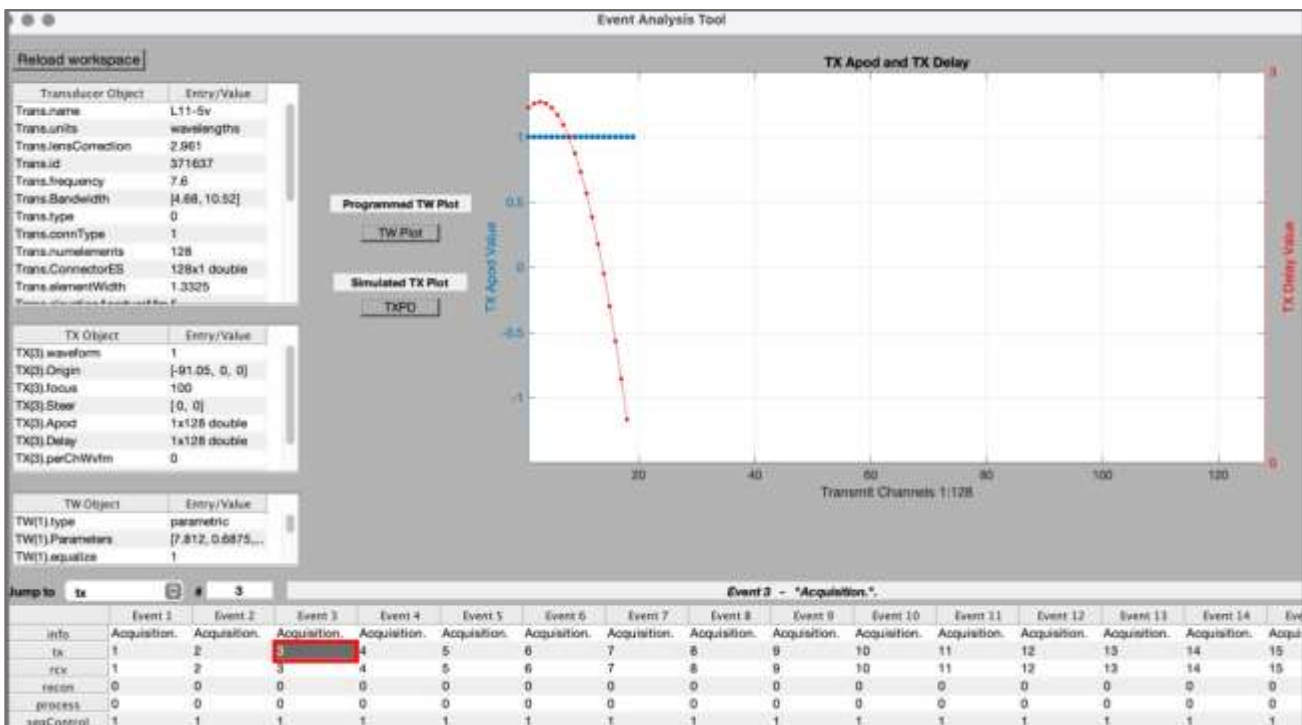


Figure 6: Event Analysis tool displays the content of structures indexed by a selected event

In addition, the Event Analysis Tool provides debugging capabilities—there are six scripts available (in the *EventAnalysisTool/Examples subfolder*) that help illustrate how to utilize the tool to find bugs in the script. The corresponding corrected scripts are also provided.

Table 1: Error Scripts and description of the corresponding warning messages

Error Script	Warning Message
ErrorScript1	Warning message if any of the events has an empty field
ErrorScript2	Each defined TX and Receive should be used in the sequence
ErrorScript3	Consistency of <code>frameNum</code> or <code>BufferNum</code> in <code>Recon/ReconInfo</code> and <code>transferToHost</code>
ErrorScript4	A bug in the callback! The <code>endDepth</code> of all Receive used for one Recon should be identical
ErrorScript5	Each Receive should not have the same <code>bufnum</code> , <code>framenum</code> , <code>acqNum</code> , and <code>mode</code> ; otherwise, the data is overwritten to the same memory location.
ErrorScript6	Each <code>transferToHost</code> must be unique in the event sequence.

Typically, Recon and ReconInfo can be difficult to inspect; if a script crashes MATLAB, the user should use Event Analysis Tool to inspect the Recon/ReconInfo setting.

A more detailed explanation (using example scripts) of the capabilities of the Event Analysis Tool is provided in the [Application Note - Event Analysis Tool](#).

5. Simple Impedance Tool

The Simple Impedance Tool can be used to visualize a plot depicting the probe impedance of each of the transducer probe element signals. Note that the tool requires that the Vantage hardware system be connected to the host controller.

Run the `ProbeImpedanceCheck.m` function (located in the Simple Impedance Tool folder) to obtain the impedance plot.

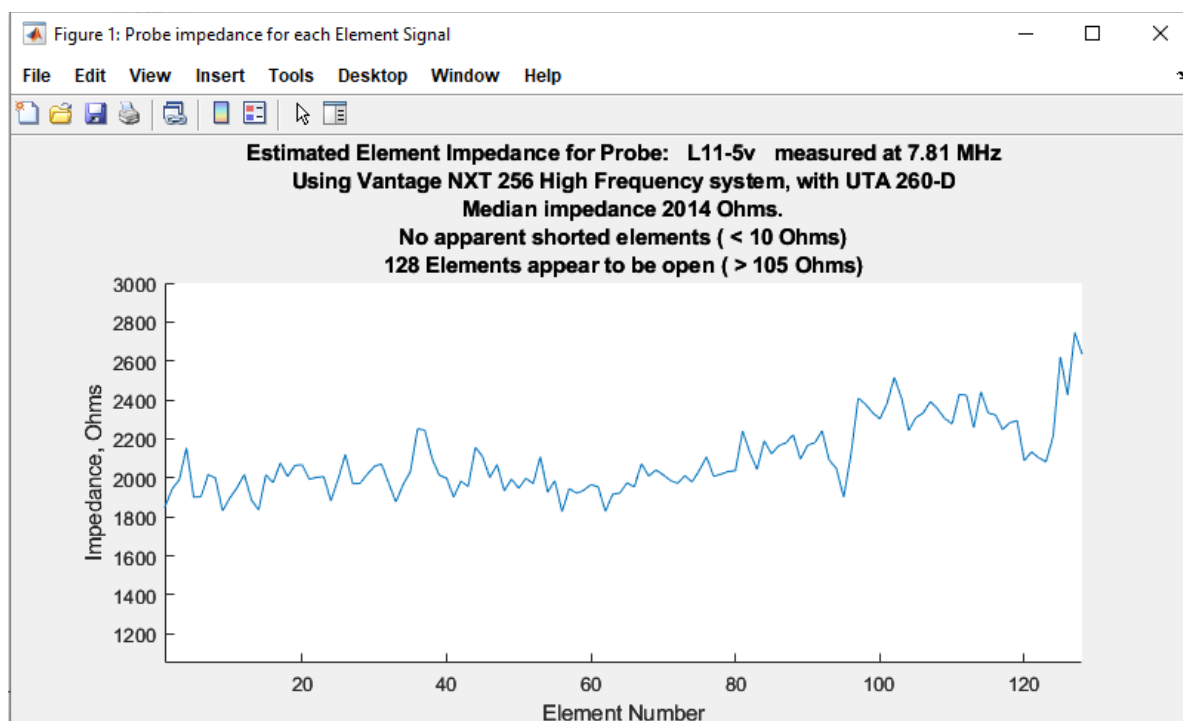


Figure 7: Simple Impedance Tool

6. Imaging Preset Design Tool

The preSetTool is used to save the configuration of the UI controls as shown in the VSX Control GUI for most example scripts, PTool, ColorMapTool, filterTool, and some predefined UI controls using VsSlider and VsButtonGroup definition.

The user can save all UI controls (such as TGC, TGC All Gain, Digital Gain, Range, High Voltage P1, High Voltage P2, and Speed of Sound) as a preset after optimizing for a particular application and easily reuse the saved preset instead of needing to manually readjust the controls.

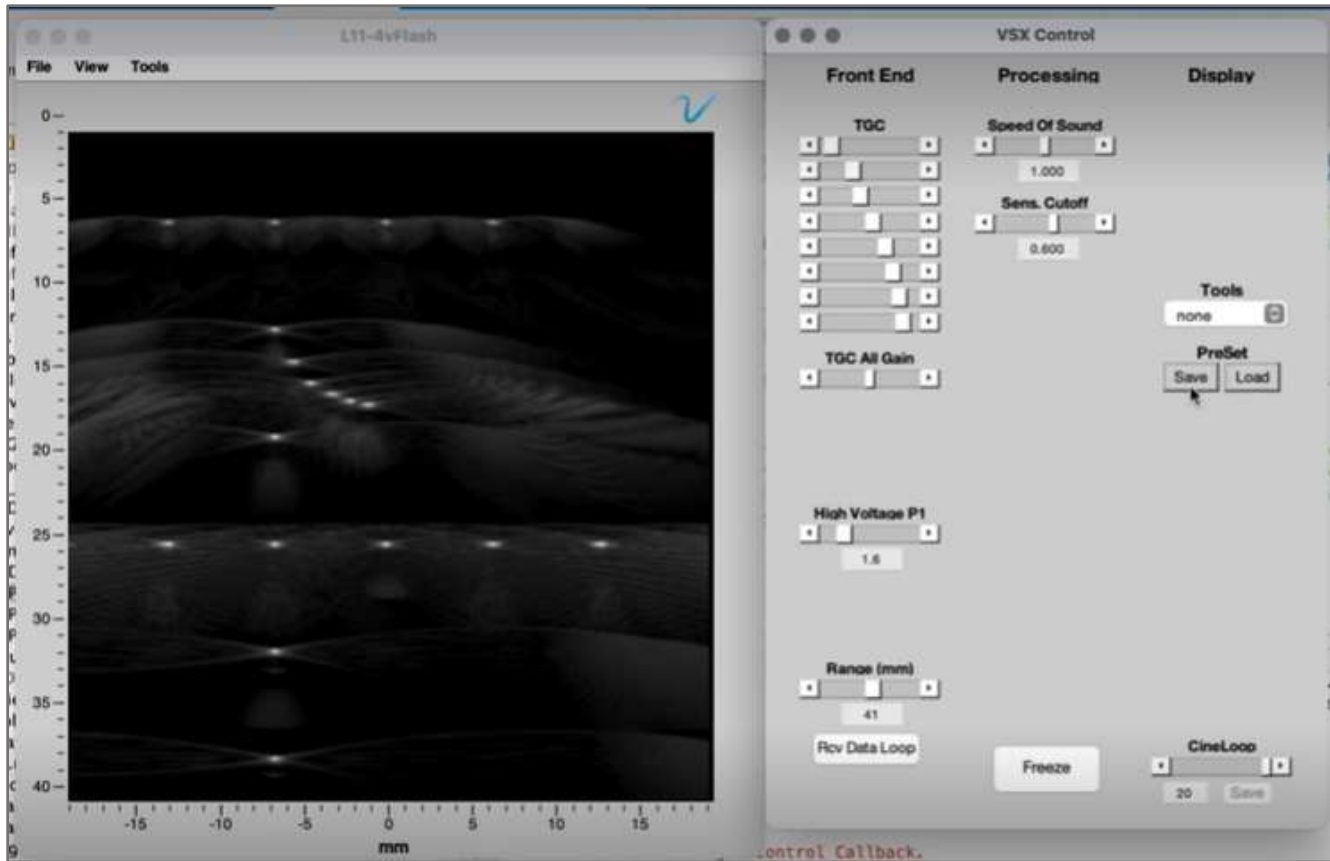


Figure 8: Preset tool

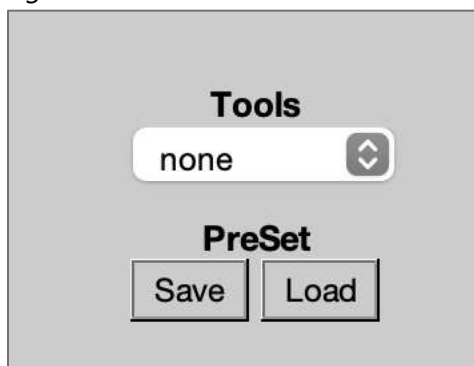


Figure 9: Save/Load options - Preset tool

The source code of the preSet function has been embedded in vsx_gui.m as nested functions; consequently, no additional folder is provided in the Tools folder. If the user-defined variables are modified by the callbacks of any UI controls, these variables should be placed in the P structure that is saved in the preset file; this ensures the correct operation of the preSetTool function.

After clicking “Save”, the filename defaults to the title of the display window. The user can rename and save all presets in a desired folder. All configurations are saved in a variable named “preSet”, within the settings file the user has defined.

The user can use “Load” to restore the settings back to what is saved in the preset file.

6.1 Important Notes

- DO NOT load the preset file before the image is displayed. MATLAB can crash if the preset file attempts to modify the variables in the workspace while the system is being initialized.
- The preSetTool is only used to save some variables in the MATLAB workspace and the status of the UI controls of the SetUp scripts provided by Verasonics. The user cannot choose the variables that can be saved. The Preset Save button saves a set of predefined variables in the workspace—such as Trans, TW, TX, speedCorrect, Receive, PData, ColorMap, customGamma, Recon, ReconInfo, Process, Doppler, P, TGC, predefined UI controls using VSlider and VsButtonGroup definition.
- Depending on the complexity of the callback functions, all settings with custom GUI controls may not be reconfigured properly.
- The preSet save only supports the predefined UI controls using VSlider and VsButtonGroup definition and does not support other custom UI controls. In addition, the Zoom and Pan in the Display panel are not saved.
- Since the SetUp script and UI controls are different between probes and applications, the correct probe name and application should be indicated in the filename. Using a preset file from a different script may cause unexpected errors or cause a software crash.
- If the callback function of a UI control changes the value or status of other UI controls, the preSetTool does not restore all parameters correctly.
- The callback functions of Doppler Mode, Compression, and Reject change the color map of the display window. The callback function of the Doppler Mode selection overwrites the Map change caused by Compression and Reject. Therefore, after loading the preset, the execution of the callback function of Compression or Reject is required to display the correct image. For instance, clicking the Reject slider and setting it back to the correct value will produce the correct image associated with the preSet.

7. Process Tool

The Process Tool or PTool, assists in modifying image processing parameters.

Use the dropdown menu in the main VSX Control GUI window to select the PTool option—a new window (with controls for modifying all of the Process image attributes) appears.

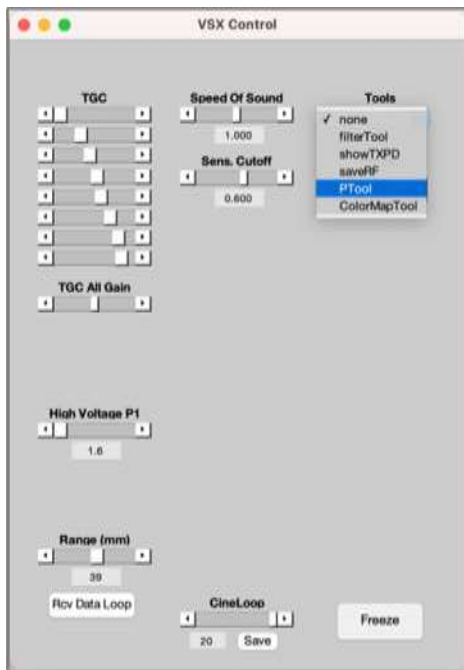


Figure 10: PTool selection from the VSX Control GUI

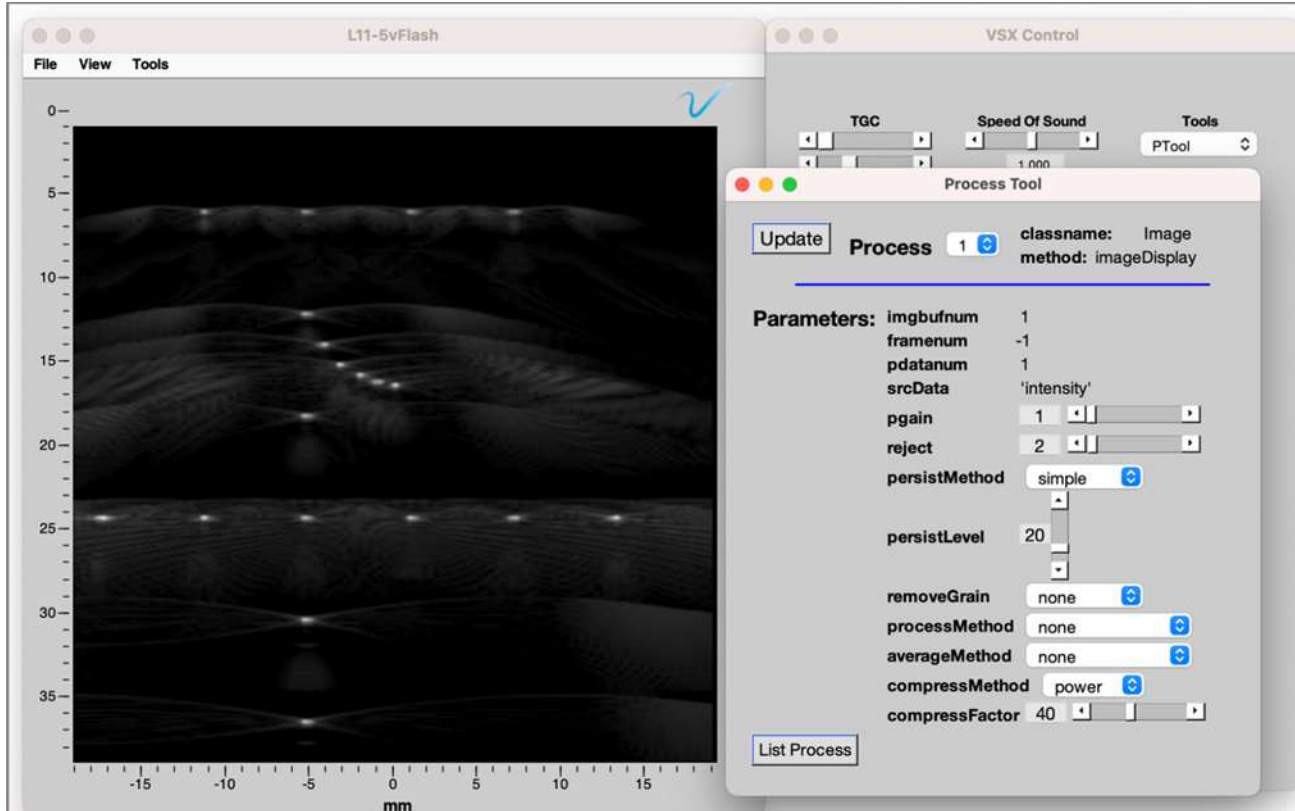


Figure 11: Process tool or PTool displaying image processing parameters

On clicking the List Process button, the Process structure (with any adjustments made to the processing attributes) is written to the MATLAB command line. The user can then copy the modified Process structure parameters into the setup script.

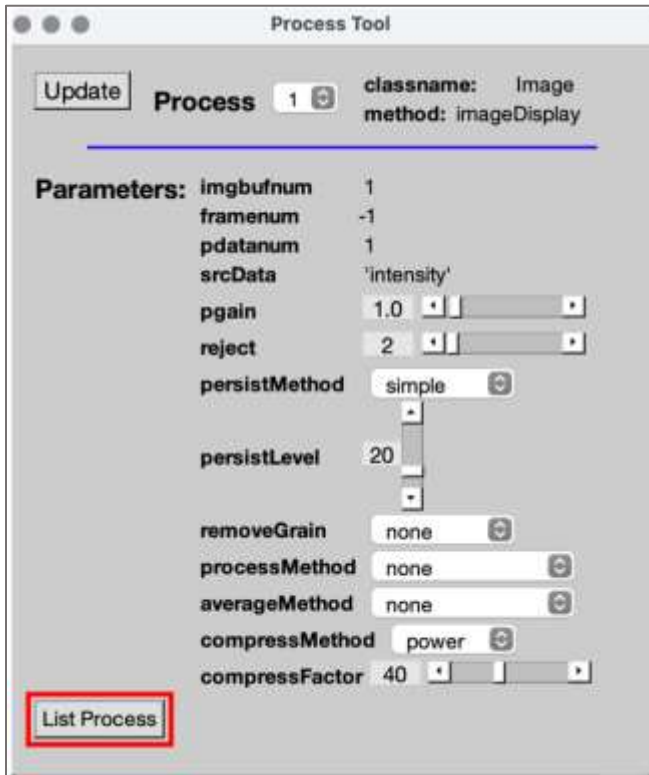


Figure 12: List Process option in the PTool

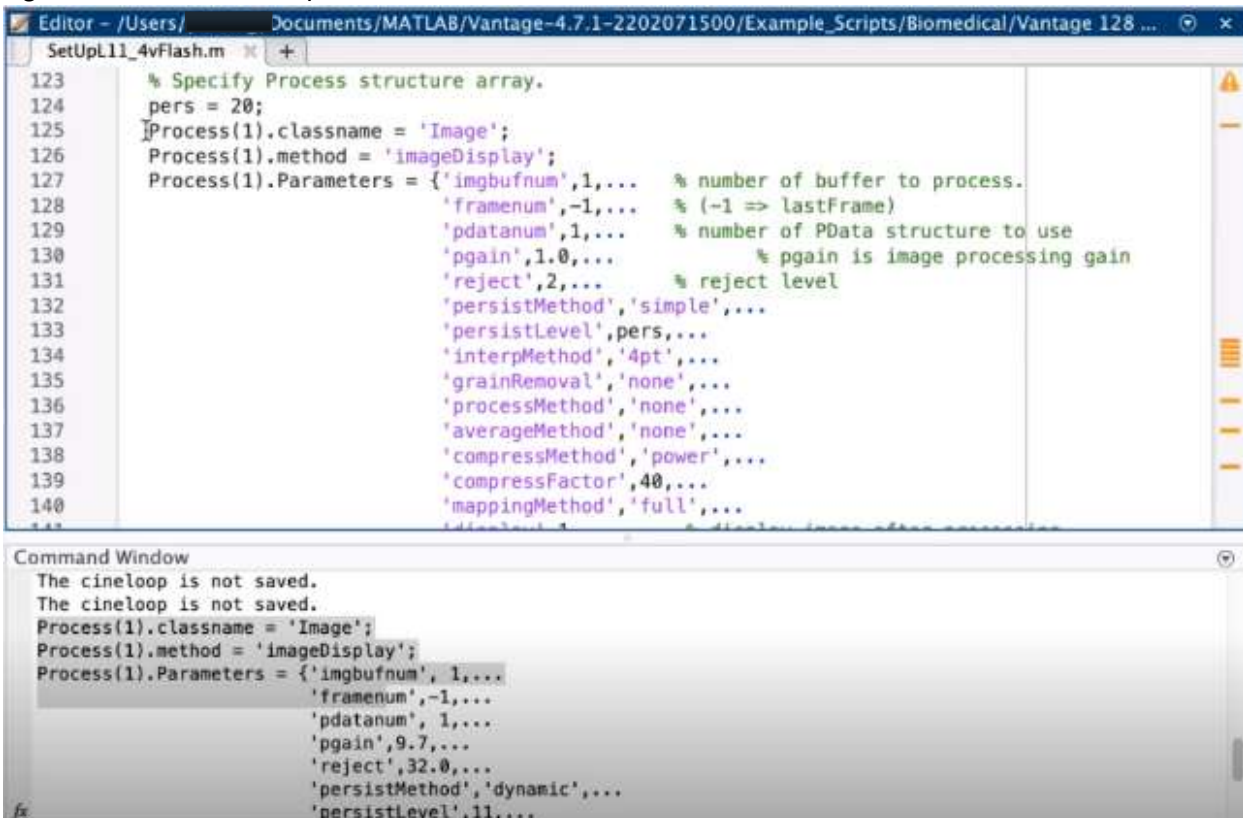


Figure 13: Process structure parameters are written to the MATLAB Command Line

8. RF Filter Design Tool

The RF Filter Design Tool is a utility that assists with designing and reviewing the digital filters provided in the receive data path for each channel of the Vantage hardware system. Refer to the Vantage Sequence Programming Manual (Section 3.3: Receive Object, and the Bandwidth sampling white paper available at <http://verasonics.com/resources-downloads/>) for descriptions of different `sampleMode` and how Verasonics implements the Bandwidth Sampling method (NS200BW and BS67BW) for high-frequency transducers. In addition to the documentation, a video is provided on the Verasonics website.

9. RF Data Viewer

The showRcvData is a stand-alone utility used to view RF data in the RcvData buffer using a grayscale image and line plots.

To view the gray scale image and line plots of RF data in the RcvData buffer-

1. Click on the showRcvData.m function (located in the *Tools/RFDataViewer_Tool* folder).
2. Run the showRcvData.m function to view the plots.

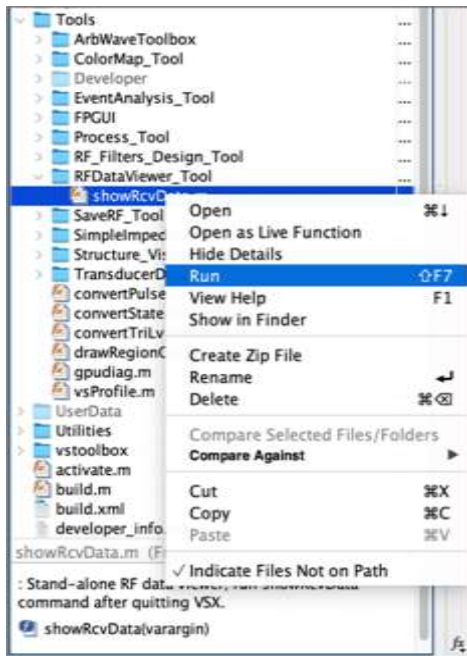


Figure 14: showRcvData utility

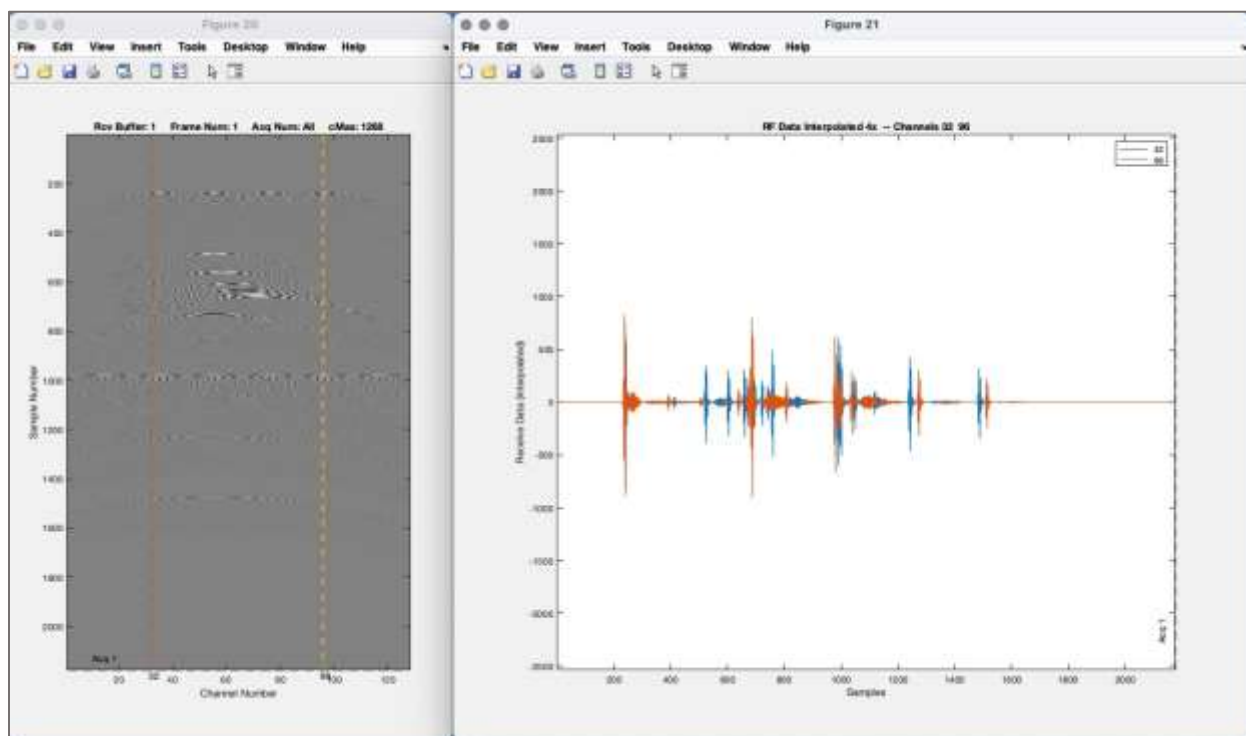


Figure 15: View RF data in the RcvData buffer using a grayscale image and line plots

To view the plots with default settings, call the `showRcvData` function without arguments.

To view the plots with the variable settings (as defined in Table 2), call the `showRcvData('variableName', numericalValue)` with arguments.

Example:

```
showRcvData('bufferNum', 1, 'frameNum', 2)
```

Table 2: Variable names and description

Variable Names	Default	Description
bufferNum	1	Receive Buffer index
frameNum	1	Frame number
acqNum	0	Acquisition number: =0 All =N Nth acquisition
chNums	[32 96]	channel numbers to plot as lines (=0 skips all line plotting) (values for 128 channels)
interpF	4	interpolation upsampling factor for the RF data line plots (=1 for no interpolation)
plotSamplePoints	0	=1 plots '+' at each true data point, and 'o' at each interpolated sample, w/o line.
cMax	500	The absolute value of data above which color is saturated black or white in the 2D image and scale line plots to $[-2*cMax \ 2*cMax]$.
ordering	'channel'	'channel': plot data in channel order, exactly as stored in the RcvData structure. 'element': plot data in element order, using Trans.Connector vector to unscramble the RF data.

By default, the program plots an entire frame and marks the boundaries between acquisition events in the frame. The zoom feature is turned on by default; this allows a user to use the figure's rubber band box zoom control to examine details of the data for a smaller region within a frame. If an acquisition number is provided, that acquisition (for frame=frameNum) is plotted. The data can also be interpolated by interpF to make the image smoother. If ordering is by the element, then chNums is used to represent element number(s) to plot. This version of the function is intended to be used from the MATLAB interface after running VSX. A version suitable for use as an external function, which requires that the only input argument be a local RF data buffer variable name, can be developed from this function.

10. Save RF Tool

The Save RF tool allows the user to save RF data to a desired location for post-processing.

To save RFdata-

1. Run the script in visualization mode.
2. On the VSX Control GUI:
 - a. Click the Freeze button.
 - b. Select the Save RF Tool option from the Tools dropdown menu.
3. After selecting the Save RF Data Tool, save the RF data as a MAT file into a desired location for post-processing.

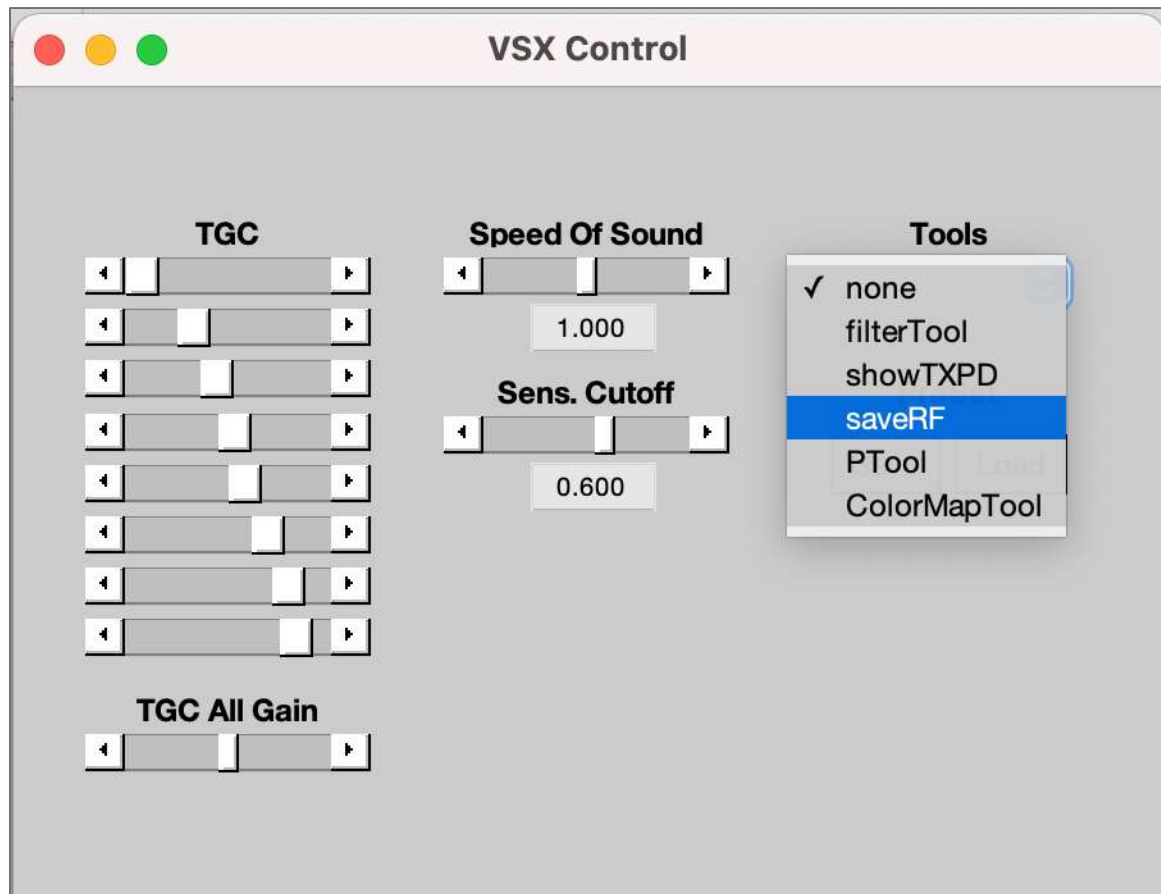


Figure 16: Save RF Tool

11. Structure Visualization

This folder collects all tools used for visualizing structures, including `ElementToChannelMapping_Tools`, `ShowGeometry_Tools`, `ShowTrans_Tools`, and `showTXPD_tool`.

11.1 Element To Channel Mapping Tool

`ElementToChannelMapping` is a useful tool to help understand the actual mapping between transducer element numbers, system channel numbers, entries in the TX/Receive Apod arrays, and pin numbers at the transducer connector.

The `Trans.Connector` array (or the selected column of `Trans.HVMux.Aperture` for HV Mux probes) defines the mapping from individual transducer elements to the associated channel in the Vantage hardware System for the selected connector or connectors. These channels are identified in the *Sequence Programming* manual as "I/O channels", to distinguish them from physical hardware channels (in some cases, such as when using a single HDI connector on a Vantage 256 system, the number of I/O channels may be less than the number of physical hardware channels). There is always an exact 1-to-1 mapping between I/O channels and columns of the Receive Data buffer in the MATLAB workspace; the number of columns in the Receive buffer definition (`Resource.RcvBuffer.colsPerFrame`) is always equal to the number of I/O channels for the selected connector(s).

In summary, the length of the `Trans.Connector` array is always equal to the number of transducer elements identified in the `Trans` structure. The *N*th entry in that array defines the connection for element *N* of the transducer definition, and the value of that entry identifies the I/O channel number to the element is connected to. A value of zero means that element is not connected at all and cannot be used by the system. All of this remains exactly the same for the UTA system configuration, for all UTA adapter modules.

Defining the `Trans.Connector` array for a user-supplied transducer

If the user defines a script for a probe recognized by the Vantage software and supported through the `computeTrans` function, all of the element-to-channel mappings are automatically handled.

If the user connects test equipment to a particular individual element signal at the probe connector, they need to find the associated connector pin number. Note that the `Trans.Connector` array defines the element-to-channel mapping but does not specify which physical connector pin is associated with each element.

If the user connects a transducer that is not supported by `computeTrans` to the system, they need to define the `Trans.Connector` array.

These issues were complicated enough on the original system; in the UTA system they get even more confusing since there are multiple UTA adapters supporting multiple connector types to choose from, and each UTA adapter and connector has a unique pin assignment at the connector and a unique mapping of those pins to system channels.

The Vantage software includes some utilities to make the element-to-channel-to-connector pin mapping easier to understand:

11.1.1 showEL

After running any script with VSX, and then exiting VSX, the MATLAB workspace still contains all of the structures that were used while the script was running.

Type the command `showEL` at the command prompt, and a listing of all transducer elements is displayed in the MATLAB Command Window, showing the system I/O channel number, physical connector pin number, and Apod array entry number (applies to both Receive.Apod and TX.Apod) associated with that element.

Enter the command with an argument, i.e. `showEL(7)`, and the mapping only for the identified element is displayed. While working with a script, this utility makes it quick and easy to find the connector pin number and I/O channel number (which is also the column number in the Receive Buffer for the associated receive data) for that element.

```
>> showEL

ans =

128x1 cell array

{'EL 001   CH 001   pin J1-U5   APOD(001)'}
{'EL 002   CH 002   pin J1-U4   APOD(002)'}
{'EL 003   CH 003   pin J1-U3   APOD(003)'}
{'EL 004   CH 004   pin J1-U2   APOD(004)'}
{'EL 005   CH 005   pin J1-V2   APOD(005)'}
{'EL 006   CH 006   pin J1-V3   APOD(006)'}
{'EL 007   CH 007   pin J1-V4   APOD(007)'}
{'EL 008   CH 008   pin J1-V5   APOD(008)'}
{'EL 009   CH 009   pin J1-W5   APOD(009)'}
{'EL 010   CH 010   pin J1-W4   APOD(010)'}
{'EL 011   CH 011   pin J1-W3   APOD(011)'}
{'EL 012   CH 012   pin J1-W2   APOD(012)'}
{'EL 013   CH 013   pin J1-X2   APOD(013)'}
{'EL 014   CH 014   pin J1-X3   APOD(014)'}
{'EL 015   CH 015   pin J1-X4   APOD(015)'}
{'EL 016   CH 016   pin J1-X5   APOD(016)'}
{'EL 017   CH 017   pin J1-Y5   APOD(017)'}
{'EL 018   CH 018   pin J1-Y4   APOD(018)'}
{'EL 019   CH 019   pin J1-Y3   APOD(019)'}
{'EL 020   CH 020   pin J1-Y2   APOD(020)'}
{'EL 021   CH 021   pin J1-Z2   APOD(021)'}
{'EL 022   CH 022   pin J1-Z3   APOD(022)'}
{'EL 023   CH 023   pin J1-Z4   APOD(023)'}
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{'EL 028   CH 028   pin J1-a2   APOD(028)'}
{'EL 029   CH 029   pin J1-a1   APOD(029)'}
{'EL 030   CH 030   pin J1-a0   APOD(030)'}
{'EL 031   CH 031   pin J1-b5   APOD(031)'}
{'EL 032   CH 032   pin J1-b4   APOD(032)'}
{'EL 033   CH 033   pin J1-b3   APOD(033)'}
{'EL 034   CH 034   pin J1-b2   APOD(034)'}
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{'EL 037   CH 037   pin J1-c5   APOD(037)'}
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{'EL 039   CH 039   pin J1-c3   APOD(039)'}
{'EL 040   CH 040   pin J1-c2   APOD(040)'}
{'EL 041   CH 041   pin J1-c1   APOD(041)'}
{'EL 042   CH 042   pin J1-c0   APOD(042)'}
{'EL 043   CH 043   pin J1-d5   APOD(043)'}
{'EL 044   CH 044   pin J1-d4   APOD(044)'}
{'EL 045   CH 045   pin J1-d3   APOD(045)'}
{'EL 046   CH 046   pin J1-d2   APOD(046)'}
{'EL 047   CH 047   pin J1-d1   APOD(047)'}
{'EL 048   CH 048   pin J1-d0   APOD(048)'}
{'EL 049   CH 049   pin J1-e5   APOD(049)'}
{'EL 050   CH 050   pin J1-e4   APOD(050)'}
{'EL 051   CH 051   pin J1-e3   APOD(051)'}
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{'EL 054   CH 054   pin J1-e0   APOD(054)'}
{'EL 055   CH 055   pin J1-f5   APOD(055)'}
{'EL 056   CH 056   pin J1-f4   APOD(056)'}
{'EL 057   CH 057   pin J1-f3   APOD(057)'}
{'EL 058   CH 058   pin J1-f2   APOD(058)'}
{'EL 059   CH 059   pin J1-f1   APOD(059)'}
{'EL 060   CH 060   pin J1-f0   APOD(060)'}
{'EL 061   CH 061   pin J1-g5   APOD(061)'}
{'EL 062   CH 062   pin J1-g4   APOD(062)'}
{'EL 063   CH 063   pin J1-g3   APOD(063)'}
{'EL 064   CH 064   pin J1-g2   APOD(064)'}
{'EL 065   CH 065   pin J1-g1   APOD(065)'}
{'EL 066   CH 066   pin J1-g0   APOD(066)'}
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{'EL 068   CH 068   pin J1-h4   APOD(068)'}
{'EL 069   CH 069   pin J1-h3   APOD(069)'}
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{'EL 073   CH 073   pin J1-i5   APOD(073)'}
{'EL 074   CH 074   pin J1-i4   APOD(074)'}
{'EL 075   CH 075   pin J1-i3   APOD(075)'}
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{'EL 083   CH 083   pin J1-j1   APOD(083)'}
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{'EL 086   CH 086   pin J1-k4   APOD(086)'}
{'EL 087   CH 087   pin J1-k3   APOD(087)'}
{'EL 088   CH 088   pin J1-k2   APOD(088)'}
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{'EL 100   CH 100   pin J1-m2   APOD(100)'}
{'EL 101   CH 101   pin J1-m1   APOD(101)'}
{'EL 102   CH 102   pin J1-m0   APOD(102)'}
{'EL 103   CH 103   pin J1-n5   APOD(103)'}
{'EL 104   CH 104   pin J1-n4   APOD(104)'}
{'EL 105   CH 105   pin J1-n3   APOD(105)'}
{'EL 106   CH 106   pin J1-n2   APOD(106)'}
{'EL 107   CH 107   pin J1-n1   APOD(107)'}
{'EL 108   CH 108   pin J1-n0   APOD(108)'}
{'EL 109   CH 109   pin J1-o5   APOD(109)'}
{'EL 110   CH 110   pin J1-o4   APOD(110)'}
{'EL 111   CH 111   pin J1-o3   APOD(111)'}
{'EL 112   CH 112   pin J1-o2   APOD(112)'}
{'EL 113   CH 113   pin J1-o1   APOD(113)'}
{'EL 114   CH 114   pin J1-o0   APOD(114)'}
{'EL 115   CH 115   pin J1-p5   APOD(115)'}
{'EL 116   CH 116   pin J1-p4   APOD(116)'}
{'EL 117   CH 117   pin J1-p3   APOD(117)'}
{'EL 118   CH 118   pin J1-p2   APOD(118)'}
{'EL 119   CH 119   pin J1-p1   APOD(119)'}
{'EL 120   CH 120   pin J1-p0   APOD(120)'}
{'EL 121   CH 121   pin J1-q5   APOD(121)'}
{'EL 122   CH 122   pin J1-q4   APOD(122)'}
{'EL 123   CH 123   pin J1-q3   APOD(123)'}
{'EL 124   CH 124   pin J1-q2   APOD(124)'}
{'EL 125   CH 125   pin J1-q1   APOD(125)'}
{'EL 126   CH 126   pin J1-q0   APOD(126)'}
{'EL 127   CH 127   pin J1-r5   APOD(127)'}
{'EL 128   CH 128   pin J1-r4   APOD(128)'}
{'EL 129   CH 129   pin J1-r3   APOD(129)'}
{'EL 130   CH 130   pin J1-r2   APOD(130)'}
{'EL 131   CH 131   pin J1-r1   APOD(131)'}
{'EL 132   CH 132   pin J1-r0   APOD(132)'}
{'EL 133   CH 133   pin J1-s5   APOD(133)'}
{'EL 134   CH 134   pin J1-s4   APOD(134)'}
{'EL 135   CH 135   pin J1-s3   APOD(135)'}
{'EL 136   CH 136   pin J1-s2   APOD(136)'}
{'EL 137   CH 137   pin J1-s1   APOD(137)'}
{'EL 138   CH 138   pin J1-s0   APOD(138)'}
{'EL 139   CH 139   pin J1-t5   APOD(139)'}
{'EL 140   CH 140   pin J1-t4   APOD(140)'}
{'EL 141   CH 141   pin J1-t3   APOD(141)'}
{'EL 142   CH 142   pin J1-t2   APOD(142)'}
{'EL 143   CH 143   pin J1-t1   APOD(143)'}
{'EL 144   CH 144   pin J1-t0   APOD(144)'}
{'EL 145   CH 145   pin J1-u5   APOD(145)'}
{'EL 146   CH 146   pin J1-u4   APOD(146)'}
{'EL 147   CH 147   pin J1-u3   APOD(147)'}
{'EL 148   CH 148   pin J1-u2   APOD(148)'}
{'EL 149   CH 149   pin J1-u1   APOD(149)'}
{'EL 150   CH 150   pin J1-u0   APOD(150)'}
{'EL 151   CH 151   pin J1-v5   APOD(151)'}
{'EL 152   CH 152   pin J1-v4   APOD(152)'}
{'EL 153   CH 153   pin J1-v3   APOD(153)'}
{'EL 154   CH 154   pin J1-v2   APOD(154)'}
{'EL 155   CH 155   pin J1-v1   APOD(155)'}
{'EL 156   CH 156   pin J1-v0   APOD(156)'}
{'EL 157   CH 157   pin J1-w5   APOD(157)'}
{'EL 158   CH 158   pin J1-w4   APOD(158)'}
{'EL 159   CH 159   pin J1-w3   APOD(159)'}
{'EL 160   CH 160   pin J1-w2   APOD(160)'}
{'EL 161   CH 161   pin J1-w1   APOD(161)'}
{'EL 162   CH 162   pin J1-w0   APOD(162)'}
{'EL 163   CH 163   pin J1-x5   APOD(163)'}
{'EL 164   CH 164   pin J1-x4   APOD(164)'}
{'EL 165   CH 165   pin J1-x3   APOD(165)'}
{'EL 166   CH 166   pin J1-x2   APOD(166)'}
{'EL 167   CH 167   pin J1-x1   APOD(167)'}
{'EL 168   CH 168   pin J1-x0   APOD(168)'}
{'EL 169   CH 169   pin J1-y5   APOD(169)'}
{'EL 170   CH 170   pin J1-y4   APOD(170)'}
{'EL 171   CH 171   pin J1-y3   APOD(171)'}
{'EL 172   CH 172   pin J1-y2   APOD(172)'}
{'EL 173   CH 173   pin J1-y1   APOD(173)'}
{'EL 174   CH 174   pin J1-y0   APOD(174)'}
{'EL 175   CH 175   pin J1-z5   APOD(175)'}
{'EL 176   CH 176   pin J1-z4   APOD(176)'}
{'EL 177   CH 177   pin J1-z3   APOD(177)'}
{'EL 178   CH 178   pin J1-z2   APOD(178)'}
{'EL 179   CH 179   pin J1-z1   APOD(179)'}
{'EL 180   CH 180   pin J1-z0   APOD(180)'}
{'EL 181   CH 181   pin J1-aa5   APOD(181)'}
{'EL 182   CH 182   pin J1-aa4   APOD(182)'}
{'EL 183   CH 183   pin J1-aa3   APOD(183)'}
{'EL 184   CH 184   pin J1-aa2   APOD(184)'}
{'EL 185   CH 185   pin J1-aa1   APOD(185)'}
{'EL 186   CH 186   pin J1-aa0   APOD(186)'}
{'EL 187   CH 187   pin J1-ab5   APOD(187)'}
{'EL 188   CH 188   pin J1-ab4   APOD(188)'}
{'EL 189   CH 189   pin J1-ab3   APOD(189)'}
{'EL 190   CH 190   pin J1-ab2   APOD(190)'}
{'EL 191   CH 191   pin J1-ab1   APOD(191)'}
{'EL 192   CH 192   pin J1-ab0   APOD(192)'}
{'EL 193   CH 193   pin J1-ac5   APOD(193)'}
{'EL 194   CH 194   pin J1-ac4   APOD(194)'}
{'EL 195   CH 195   pin J1-ac3   APOD(195)'}
{'EL 196   CH 196   pin J1-ac2   APOD(196)'}
{'EL 197   CH 197   pin J1-ac1   APOD(197)'}
{'EL 198   CH 198   pin J1-ac0   APOD(198)'}
{'EL 199   CH 199   pin J1-ad5   APOD(199)'}
{'EL 200   CH 200   pin J1-ad4   APOD(200)'}
{'EL 201   CH 201   pin J1-ad3   APOD(201)'}
{'EL 202   CH 202   pin J1-ad2   APOD(202)'}
{'EL 203   CH 203   pin J1-ad1   APOD(203)'}
{'EL 204   CH 204   pin J1-ad0   APOD(204)'}
{'EL 205   CH 205   pin J1-ae5   APOD(205)'}
{'EL 206   CH 206   pin J1-ae4   APOD(206)'}
{'EL 207   CH 207   pin J1-ae3   APOD(207)'}
{'EL 208   CH 208   pin J1-ae2   APOD(208)'}
{'EL 209   CH 209   pin J1-ae1   APOD(209)'}
{'EL 210   CH 210   pin J1-ae0   APOD(210)'}
{'EL 211   CH 211   pin J1-af5   APOD(211)'}
{'EL 212   CH 212   pin J1-af4   APOD(212)'}
{'EL 213   CH 213   pin J1-af3   APOD(213)'}
{'EL 214   CH 214   pin J1-af2   APOD(214)'}
{'EL 215   CH 215   pin J1-af1   APOD(215)'}
{'EL 216   CH 216   pin J1-af0   APOD(216)'}
{'EL 217   CH 217   pin J1-ag5   APOD(217)'}
{'EL 218   CH 218   pin J1-ag4   APOD(218)'}
{'EL 219   CH 219   pin J1-ag3   APOD(219)'}
{'EL 220   CH 220   pin J1-ag2   APOD(220)'}
{'EL 221   CH 221   pin J1-ag1   APOD(221)'}
{'EL 222   CH 222   pin J1-ag0   APOD(222)'}
{'EL 223   CH 223   pin J1-ah5   APOD(223)'}
{'EL 224   CH 224   pin J1-ah4   APOD(224)'}
{'EL 225   CH 225   pin J1-ah3   APOD(225)'}
{'EL 226   CH 226   pin J1-ah2   APOD(226)'}
{'EL 227   CH 227   pin J1-ah1   APOD(227)'}
{'EL 228   CH 228   pin J1-ah0   APOD(228)'}
{'EL 229   CH 229   pin J1-ai5   APOD(229)'}
{'EL 230   CH 230   pin J1-ai4   APOD(230)'}
{'EL 231   CH 231   pin J1-ai3   APOD(231)'}
{'EL 232   CH 232   pin J1-ai2   APOD(232)'}
{'EL 233   CH 233   pin J1-ai1   APOD(233)'}
{'EL 234   CH 234   pin J1-ai0   APOD(234)'}
{'EL 235   CH 235   pin J1-aj5   APOD(235)'}
{'EL 236   CH 236   pin J1-aj4   APOD(236)'}
{'EL 237   CH 237   pin J1-aj3   APOD(237)'}
{'EL 238   CH 238   pin J1-aj2   APOD(238)'}
{'EL 239   CH 239   pin J1-aj1   APOD(239)'}
{'EL 240   CH 240   pin J1-aj0   APOD(240)'}
{'EL 241   CH 241   pin J1-ak5   APOD(241)'}
{'EL 242   CH 242   pin J1-ak4   APOD(242)'}
{'EL 243   CH 243   pin J1-ak3   APOD(243)'}
{'EL 244   CH 244   pin J1-ak2   APOD(244)'}
{'EL 245   CH 245   pin J1-ak1   APOD(245)'}
{'EL 246   CH 246   pin J1-ak0   APOD(246)'}
{'EL 247   CH 247   pin J1-al5   APOD(247)'}
{'EL 248   CH 248   pin J1-al4   APOD(248)'}
{'EL 249   CH 249   pin J1-al3   APOD(249)'}
{'EL 250   CH 250   pin J1-al2   APOD(250)'}
{'EL 251   CH 251   pin J1-al1   APOD(251)'}
{'EL 252   CH 252   pin J1-al0   APOD(252)'}
{'EL 253   CH 253   pin J1-am5   APOD(253)'}
{'EL 254   CH 254   pin J1-am4   APOD(254)'}
{'EL 255   CH 255   pin J1-am3   APOD(255)'}
{'EL 256   CH 256   pin J1-am2   APOD(256)'}
{'EL 257   CH 257   pin J1-am1   APOD(257)'}
{'EL 258   CH 258   pin J1-am0   APOD(258)'}
{'EL 259   CH 259   pin J1-an5   APOD(259)'}
{'EL 260   CH 260   pin J1-an4   APOD(260)'}
{'EL 261   CH 261   pin J1-an3   APOD(261)'}
{'EL 262   CH 262   pin J1-an2   APOD(262)'}
{'EL 263   CH 263   pin J1-an1   APOD(263)'}
{'EL 264   CH 264   pin J1-an0   APOD(264)'}
{'EL 265   CH 265   pin J1-ao5   APOD(265)'}
{'EL 266   CH 266   pin J1-ao4   APOD(266)'}
{'EL 267   CH 267   pin J1-ao3   APOD(267)'}
{'EL 268   CH 268   pin J1-ao2   APOD(268)'}
{'EL 269   CH 269   pin J1-ao1   APOD(269)'}
{'EL 270   CH 270   pin J1-ao0   APOD(270)'}
{'EL 271   CH 271   pin J1-ap5   APOD(271)'}
{'EL 272   CH 272   pin J1-ap4   APOD(272)'}
{'EL 273   CH 273   pin J1-ap3   APOD(273)'}
{'EL 274   CH 274   pin J1-ap2   APOD(274)'}
{'EL 275   CH 275   pin J1-ap1   APOD(275)'}
{'EL 276   CH 276   pin J1-ap0   APOD(276)'}
{'EL 277   CH 277   pin J1-aq5   APOD(277)'}
{'EL 278   CH 278   pin J1-aq4   APOD(278)'}
{'EL 279   CH 279   pin J1-aq3   APOD(279)'}
{'EL 280   CH 280   pin J1-aq2   APOD(280)'}
{'EL 281   CH 281   pin J1-aq1   APOD(281)'}
{'EL 282   CH 282   pin J1-aq0   APOD(282)'}
{'EL 283   CH 283   pin J1-ar5   APOD(283)'}
{'EL 284   CH 284   pin J1-ar4   APOD(284)'}
{'EL 285   CH 285   pin J1-ar3   APOD(285)'}
{'EL 286   CH 286   pin J1-ar2   APOD(286)'}
{'EL 287   CH 287   pin J1-ar1   APOD(287)'}
{'EL 288   CH 288   pin J1-ar0   APOD(288)'}
{'EL 289   CH 289   pin J1-as5   APOD(289)'}
{'EL 290   CH 290   pin J1-as4   APOD(290)'}
{'EL 291   CH 291   pin J1-as3   APOD(291)'}
{'EL 292   CH 292   pin J1-as2   APOD(292)'}
{'EL 293   CH 293   pin J1-as1   APOD(293)'}
{'EL 294   CH 294   pin J1-as0   APOD(294)'}
{'EL 295   CH 295   pin J1-at5   APOD(295)'}
{'EL 296   CH 296   pin J1-at4   APOD(296)'}
{'EL 297   CH 297   pin J1-at3   APOD(297)'}
{'EL 298   CH 298   pin J1-at2   APOD(298)'}
{'EL 299   CH 299   pin J1-at1   APOD(299)'}
{'EL 300   CH 300   pin J1-at0   APOD(300)'}
{'EL 301   CH 301   pin J1-au5   APOD(301)'}
{'EL 302   CH 302   pin J1-au4   APOD(302)'}
{'EL 303   CH 303   pin J1-au3   APOD(303)'}
{'EL 304   CH 304   pin J1-au2   APOD(304)'}
{'EL 305   CH 305   pin J1-au1   APOD(305)'}
{'EL 306   CH 306   pin J1-au0   APOD(306)'}
{'EL 307   CH 307   pin J1-av5   APOD(307)'}
{'EL 308   CH 308   pin J1-av4   APOD(308)'}
{'EL 309   CH 309   pin J1-av3   APOD(309)'}
{'EL 310   CH 310   pin J1-av2   APOD(310)'}
{'EL 311   CH 311   pin J1-av1   APOD(311)'}
{'EL 312   CH 312   pin J1-av0   APOD(312)'}
{'EL 313   CH 313   pin J1-aw5   APOD(313)'}
{'EL 314   CH 314   pin J1-aw4   APOD(314)'}
{'EL 315   CH 315   pin J1-aw3   APOD(315)'}
{'EL 316   CH 316   pin J1-aw2   APOD(316)'}
{'EL 317   CH 317   pin J1-aw1   APOD(317)'}
{'EL 318   CH 318   pin J1-aw0   APOD(318)'}
{'EL 319   CH 319   pin J1-ax5   APOD(319)'}
{'EL 320   CH 320   pin J1-ax4   APOD(320)'}
{'EL 321   CH 321   pin J1-ax3   APOD(321)'}
{'EL 322   CH 322   pin J1-ax2   APOD(322)'}
{'EL 323   CH 323   pin J1-ax1   APOD(323)'}
{'EL 324   CH 324   pin J1-ax0   APOD(324)'}
{'EL 325   CH 325   pin J1-ay5   APOD(325)'}
{'EL 326   CH 326   pin J1-ay4   APOD(326)'}
{'EL 327   CH 327   pin J1-ay3   APOD(327)'}
{'EL 328   CH 328   pin J1-ay2   APOD(328)'}
{'EL 329   CH 329   pin J1-ay1   APOD(329)'}
{'EL 330   CH 330   pin J1-ay0   APOD(330)'}
{'EL 331   CH 331   pin J1-az5   APOD(331)'}
{'EL 332   CH 332   pin J1-az4   APOD(332)'}
{'EL 333   CH 333   pin J1-az3   APOD(333)'}
{'EL 334   CH 334   pin J1-az2   APOD(334)'}
{'EL 335   CH 335   pin J1-az1   APOD(335)'}
{'EL 336   CH 336   pin J1-az0   APOD(336)'}
{'EL 337   CH 337   pin J1-ba5   APOD(337)'}
{'EL 338   CH 338   pin J1-ba4   APOD(338)'}
{'EL 339   CH 339   pin J1-ba3   APOD(339)'}
{'EL 340   CH 340   pin J1-ba2   APOD(340)'}
{'EL 341   CH 341   pin J1-ba1   APOD(341)'}
{'EL 342   CH 342   pin J1-ba0   APOD(342)'}
{'EL 343   CH 343   pin J1-bb5   APOD(343)'}
{'EL 344   CH 344   pin J1-bb4   APOD(344)'}
{'EL 345   CH 345   pin J1-bb3   APOD(345)'}
{'EL 346   CH 346   pin J1-bb2   APOD(346)'}
{'EL 347   CH 347   pin J1-bb1   APOD(347)'}
{'EL 348   CH 348   pin J1-bb0   APOD(348)'}
{'EL 349   CH 349   pin J1-bc5   APOD(349)'}
{'EL 350   CH 350   pin J1-bc4   APOD(350)'}
{'EL 351   CH 351   pin J1-bc3   APOD(351)'}
{'EL 352   CH 352   pin J1-bc2   APOD(352)'}
{'EL 353   CH 353   pin J1-bc1   APOD(353)'}
{'EL 354   CH 354   pin J1-bc0   APOD(354)'}
{'EL 355   CH 355   pin J1-bd5   APOD(355)'}
{'EL 356   CH 356   pin J1-bd4   APOD(356)'}
{'EL 357   CH 357   pin J1-bd3   APOD(357)'}
{'EL 358   CH 358   pin J1-bd2   APOD(358)'}
{'EL 359   CH 359   pin J1-bd1   APOD(359)'}
{'EL 360   CH 360   pin J1-bd0   APOD(360)'}
{'EL 361   CH 361   pin J1-be5   APOD(361)'}
{'EL 362   CH 362   pin J1-be4   APOD(362)'}
{'EL 363   CH 363   pin J1-be3   APOD(363)'}
{'EL 364   CH 364   pin J1-be2   APOD(364)'}
{'EL 365   CH 365   pin J1-be1   APOD(365)'}
{'EL 366   CH 366   pin J1-be0   APOD(366)'}
{'EL 367   CH 367   pin J1-bf5   APOD(367)'}
{'EL 368   CH 368   pin J1-bf4   APOD(368)'}
{'EL 369   CH 369   pin J1-bf3   APOD(369)'}
{'EL 370   CH 370   pin J1-bf2   APOD(370)'}
{'EL 371   CH 371   pin J1-bf1   APOD(371)'}
{'EL 372   CH 372   pin J1-bf0   APOD(372)'}
{'EL 373   CH 373   pin J1-bg5   APOD(373)'}
{'EL 374   CH 374   pin J1-bg4   APOD(374)'}
{'EL 375   CH 375   pin J1-bg3   APOD(375)'}
{'EL 376   CH 376   pin J1-bg2   APOD(376)'}
{'EL 377   CH 377   pin J1-bg1   APOD(377)'}
{'EL 378   CH 378   pin J1-bg0   APOD(378)'}
{'EL 379   CH 379   pin J1-bh5   APOD(379)'}
{'EL 380   CH 380   pin J1-bh4   APOD(380)'}
{'EL 381   CH 381   pin J1-bh3   APOD(381)'}
{'EL 382   CH 382   pin J1-bh2   APOD(382)'}
{'EL 383   CH 383   pin J1-bh1   APOD(383)'}
{'EL 384   CH 384   pin J1-bh0   APOD(384)'}
{'EL 385   CH 385   pin J1-bi5   APOD(385)'}
{'EL 386   CH 386   pin J1-bi4   APOD(386)'}
{'EL 387   CH 387   pin J1-bi3   APOD(387)'}
{'EL 388   CH 388   pin J1-bi2   APOD(388)'}
{'EL 389   CH 389   pin J1-bi1   APOD(389)'}
{'EL 390   CH 390   pin J1-bi0   APOD(390)'}
{'EL 391   CH 391   pin J1-bj5   APOD(391)'}
{'EL 392   CH 392   pin J1-bj4   APOD(392)'}
{'EL 393   CH 393   pin J1-bj3   APOD(393)'}
{'EL 394   CH 394   pin J1-bj2   APOD(394)'}
{'EL 395   CH 395   pin J1-bj1   APOD(395)'}
{'EL 396   CH 396   pin J1-bj0   APOD(396)'}
{'EL 397   CH 397   pin J1-bk5   APOD(397)'}
{'EL 398   CH 398   pin J1-bk4   APOD(398)'}
{'EL 399   CH 399   pin J1-bk3   APOD(399)'}
{'EL 400   CH 400   pin J1-bk2   APOD(400)'}
{'EL 401   CH 401   pin J1-bk1   APOD(401)'}
{'EL 402   CH 402   pin J1-bk0   APOD(402)'}
{'EL 403   CH 403   pin J1-bl5   APOD(403)'}
{'EL 404   CH 404   pin J1-bl4   APOD(404)'}
{'EL 405   CH 405   pin J1-bl3   APOD(405)'}
{'EL 406   CH 406   pin J1-bl2   APOD(406)'}
{'EL 407   CH 407   pin J1-bl1   APOD(407)'}
{'EL
```

11.1.2 showCH and showAPOD

These utilities are identical to `showEL` as described above and display the same information, except that the listing is sorted by I/O channel numbers or Apod array entries instead of being sorted by elements.

```
>> showCH

ans =

128x1 cell array

{'CH 001' 'EL 001' 'pin J1-U5' 'APOD(001)'}
{'CH 002' 'EL 002' 'pin J1-U4' 'APOD(002)'}
{'CH 003' 'EL 003' 'pin J1-U3' 'APOD(003)'}
{'CH 004' 'EL 004' 'pin J1-U2' 'APOD(004)'}
{'CH 005' 'EL 005' 'pin J1-V2' 'APOD(005)'}
{'CH 006' 'EL 006' 'pin J1-V3' 'APOD(006)'}
{'CH 007' 'EL 007' 'pin J1-V4' 'APOD(007)'}
{'CH 008' 'EL 008' 'pin J1-V5' 'APOD(008)'}
{'CH 009' 'EL 009' 'pin J1-W5' 'APOD(009)'}
{'CH 010' 'EL 010' 'pin J1-W4' 'APOD(010)'}
{'CH 011' 'EL 011' 'pin J1-W3' 'APOD(011)'}
{'CH 012' 'EL 012' 'pin J1-W2' 'APOD(012)'}
{'CH 013' 'EL 013' 'pin J1-X2' 'APOD(013)'}
{'CH 014' 'EL 014' 'pin J1-X3' 'APOD(014)'}
{'CH 015' 'EL 015' 'pin J1-X4' 'APOD(015)'}
{'CH 016' 'EL 016' 'pin J1-X5' 'APOD(016)'}
{'CH 017' 'EL 017' 'pin J1-Y5' 'APOD(017)'}
{'CH 018' 'EL 018' 'pin J1-Y4' 'APOD(018)'}

```

Figure 18: showCH utility

```
>> showAPOD

ans =

128x1 cell array

{'APOD(001)    EL 001    CH 001    pin J1-U5'}
{'APOD(002)    EL 002    CH 002    pin J1-U4'}
{'APOD(003)    EL 003    CH 003    pin J1-U3'}
{'APOD(004)    EL 004    CH 004    pin J1-U2'}
{'APOD(005)    EL 005    CH 005    pin J1-V2'}
{'APOD(006)    EL 006    CH 006    pin J1-V3'}
{'APOD(007)    EL 007    CH 007    pin J1-V4'}
{'APOD(008)    EL 008    CH 008    pin J1-V5'}
{'APOD(009)    EL 009    CH 009    pin J1-W5'}
{'APOD(010)    EL 010    CH 010    pin J1-W4'}
{'APOD(011)    EL 011    CH 011    pin J1-W3'}
{'APOD(012)    EL 012    CH 012    pin J1-W2'}
{'APOD(013)    EL 013    CH 013    pin J1-X2'}
{'APOD(014)    EL 014    CH 014    pin J1-X3'}
{'APOD(015)    EL 015    CH 015    pin J1-X4'}
{'APOD(016)    EL 016    CH 016    pin J1-X5'}
{'APOD(017)    EL 017    CH 017    pin J1-Y5'}
{'APOD(018)    EL 018    CH 018    pin J1-Y4'}
{'APOD(019)    EL 019    CH 019    pin J1-Y3'}
{'APOD(020)    EL 020    CH 020    pin J1-Y2'}
```

Figure 19: showAPOD utility

To define the `Trans.Connector` array for a probe that is not recognized by `computeTrans`, the user must first determine (from the probe design documentation) to which physical connector pins each of the elements is connected to. Then the user can use one of the utilities described above to find the mapping from each connector pin to the system I/O channel; this lets the user put the correct channel number in each entry of the `Trans.Connector` array. Refer to the [Sequence Programming Manual](#) for additional information on integrating custom transducers into the system.

12. Show Geometry Tools

The `showGeometry` tool plots three structures in the Verasonics coordinate system—this includes `MediaPts`, `PData`, and `Trans` in three dimensions. If no argument is provided, a new figure is created. The user is encouraged to use the MATLAB window menu to rotate and zoom through the plot as desired. The `showPData` tool plots the `PData` structures as shaded patches in 3D.

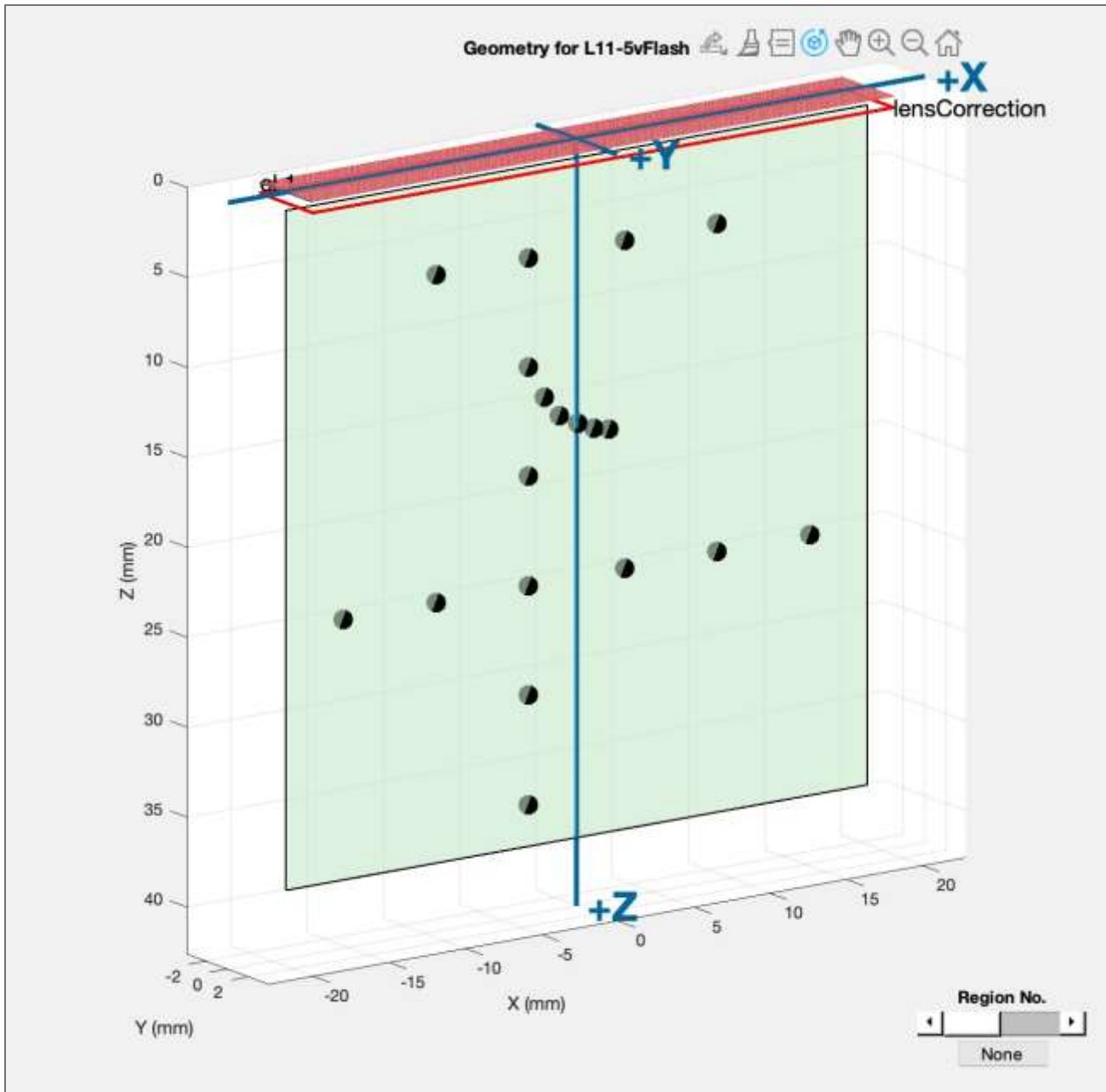


Figure 20: `showGeometry` tool

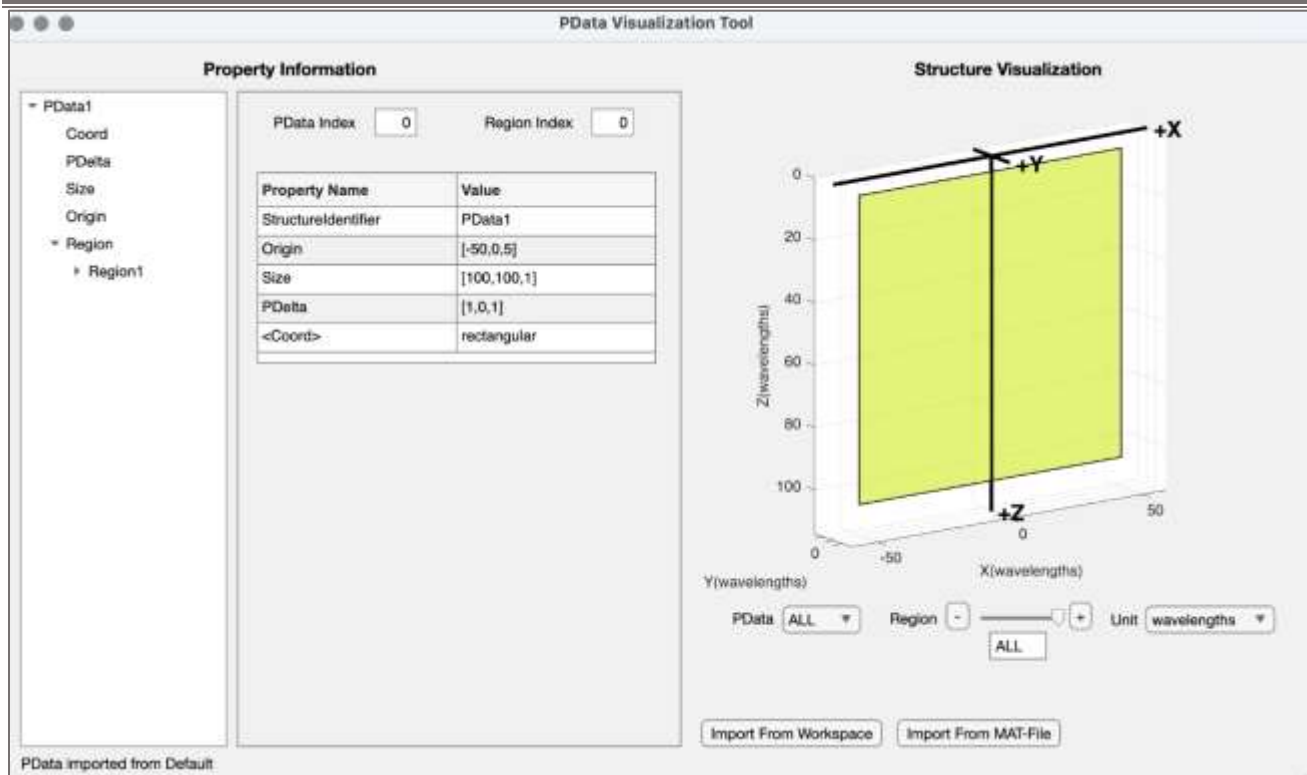


Figure 21: PData tool

13. Show Transducer Tool

showTrans provides a graphical representation of the following:

- The element sizes and locations by drawing rectangular patches in the Verasonics coordinate system. The array is placed at $Z=0$ but draws a rectangle at an effective depth resulting from the lens correction.
- The impulse responses of the transducer are provided or calculated.
- The impedance of the transducer.

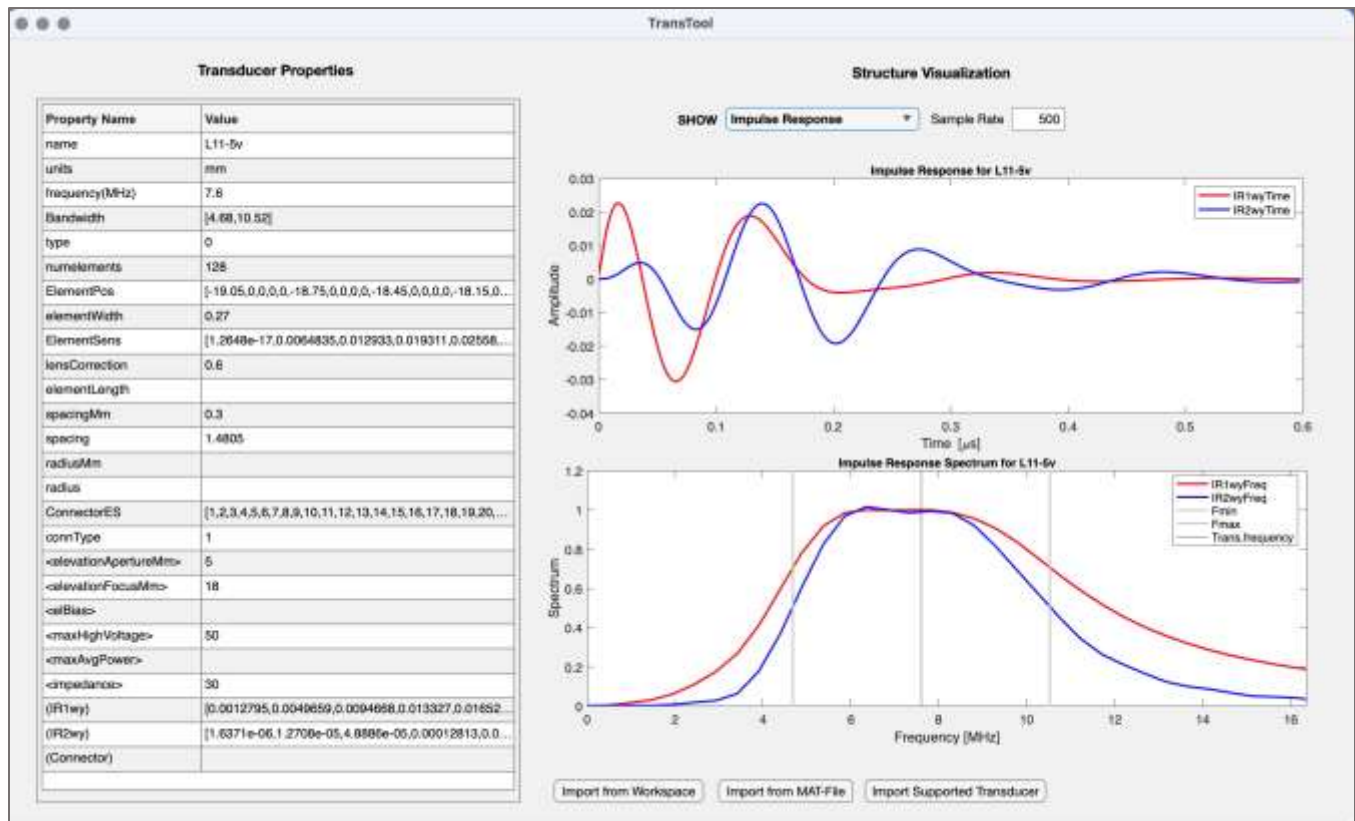


Figure 22: showTrans tool plot

14. Show TXPD Tools

When a TX structure is present in the MATLAB workspace, invoking the showTXPD tool produces and displays beam plots of the transmit beam that are produced by each individual TX structure. Refer to the [Application Note – showTXPD tool](#) on the Verasonics Community Portal.

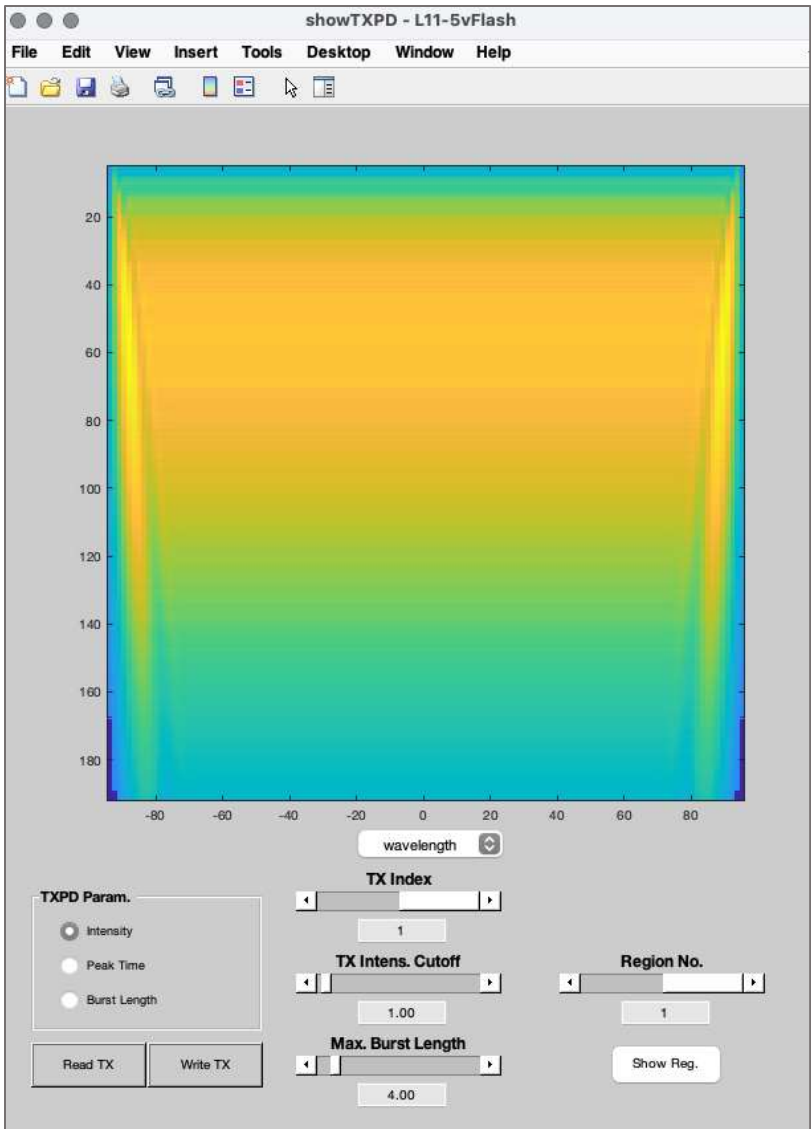


Figure 23: showTXPD tool

Table 3 provides descriptions of the GUI controls in the showTXPD tool.

Table 3: GUI controls in showTXPD tool

GUI Controls	Description
Intensity	Intensity of Transmit beam at pixel locations
Peak time	Time of arrival of peak intensity of transmit pulse at pixels
Burst Length	Length of transmit burst at pixel locations
Read TX	Reads the TX structure from the MATLAB workspace
Write TX	Writes the application’s TX structure to the MATLAB workspace

15. Show Hardware Memory Usage Utility Function

showHardwareMemoryUsage is a utility function for displaying the actual memory usage in the hardware system after a user script has been initialized and run by VSX - runAcq. This utility queries the actual hardware event sequence that has been programmed into the hardware system; consequently, it cannot be used if a hardware system is not present and the script was initialized for simulate-mode operation only.

USAGE: Simply type "showHardwareMemoryUsage" at the MATLAB command prompt. This function has no input or output arguments. The resulting memory usage report is displayed in the MATLAB Command Window.

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